



Code

MFEM Workshop 2026

Particle-in-cell in MFEM

Rushan Zhang¹ (rzhangbq@gatech.edu), Joseph Signorelli², Ketan Mittal³, Tzanio Kolev³, Qi Tang¹

¹Georgia Institute of Technology, ²University of Illinois Urbana-Champaign, ³Lawrence Livermore National Laboratory

Highlights

- Electrostatic **PIC** on MFEM particle tracing: particles deposit charge, the mesh field pushes them back
- Compatible gradient **CG** $\xrightarrow{\nabla_h}$ **ND** and **OrthoSolver** for periodic Poisson
- **Bi-harmonic-heat** shape function: a mesh-compatible low-pass filter applied by solving $u_t + \kappa \Delta^2 u = 0$



Code

One PIC time step in MFEM

particles $\xrightarrow{\text{deposit}}$ field $\xrightarrow{\text{push}}$ particles

1. Deposit charge.

For $\varphi \in H^1(\Omega)$, assemble $e \sum_p \varphi(\mathbf{x}_p)$.

2. Solve Poisson.

OrthoSolver removes the periodic nullspace.

$$\epsilon_0 \langle \nabla \varphi, \nabla \phi \rangle = e \sum_p \varphi(\mathbf{x}_p) - en_0 \int_{\Omega} \varphi.$$

3. Compute field.

$\langle \mathbf{v}, \mathbf{E} + \nabla \phi \rangle = 0$ in $H(\text{curl})$.

4. Gather field.

Interpolate $\mathbf{E}$ at each $\mathbf{x}_p$.

5. Leapfrog push.

$$\begin{aligned} \mathbf{p}_p^{t+\frac{1}{2}\Delta t} &= \mathbf{p}_p^{t-\frac{1}{2}\Delta t} + e\mathbf{E}^t(\mathbf{x}_p^t)\Delta t, \\ \mathbf{x}_p^{t+1} &= \mathbf{x}_p^t + \mathbf{p}_p^{t+\frac{1}{2}\Delta t} \Delta t / m_p. \end{aligned}$$

6. Redistribute.

Exchange particles that cross MPI ranks.

Low-pass shape function

The shape function is the biharmonic heat kernel at time τ :

$$s_{\mathbf{x}}(\mathbf{x}) = E(\mathbf{x}, \tau), \quad \hat{E}(\mathbf{k}, \tau) = e^{-\kappa(2\pi/L)^4 \|\mathbf{k}\|^4 \tau}.$$

Applying $s_{\mathbf{x}}$ is equivalent to convolving the Dirac charge (and the potential) with that kernel:

$$\rho = s_{\mathbf{x}} * \rho_{\delta}, \quad \phi_s = s_{\mathbf{x}} * \phi, \quad \mathbf{E}_p = -\nabla \phi_s(\mathbf{x}_p).$$

Mode comparison (Ex)

— Delta Shape Function — Biharmonic Heat Shape Function

